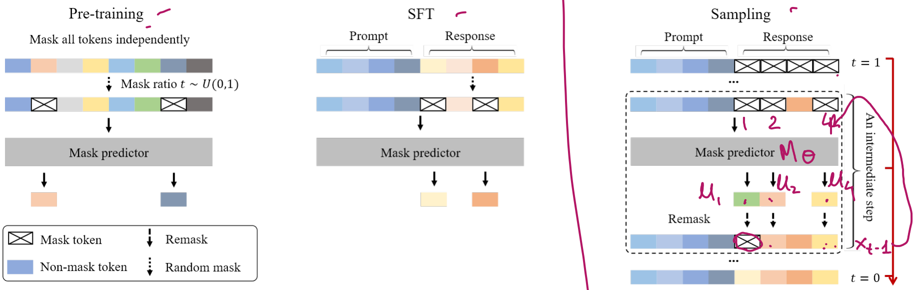
Large Language Diffusion Models Llada[2]



[2] https://ml-gsai.github.io/LLaDA-demo/

Inference in Llada

Input: Model: M_θ, Prompt: P, n: number of tokens to generate, T : maximum number of steps.

Initialize $\mathbf{x}_T = n$ masks

for t = T to 1 **do**

  Invoke $M_\theta(P, \mathbf{x}_t)$, obtain $\mu_i(.)$ for all masked positions i in $\mathbf{x}_t$

$\hat{x}_0$ $\tilde{\mathbf{x}}_{t-1}$ = sample high probability tokens from $\mu_i(.)$ for each i.

$x_{t-1} \leftarrow$ Remask $\boxed{\frac{1-\alpha_{t-1}}{1-\alpha_t}}$ fraction of tokens in $\tilde{\mathbf{x}}_{t-1}$. Several heuristics for remasking: random, lowest confidence, block auto-regressive.

end for=0